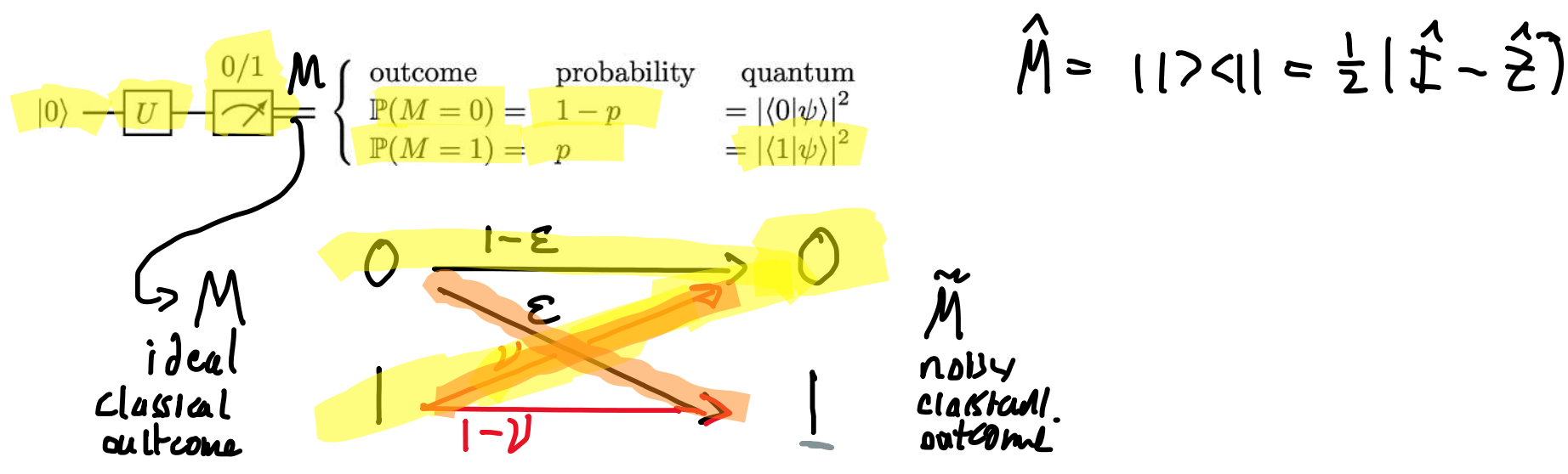


Introduction to quantum noise

Measurement error

Qiskit Global Summer School on Quantum Machine Learning

Zlatko K. Minev



$$P_M = \begin{pmatrix} P(M=0) \\ P(M=1) \end{pmatrix} = \begin{pmatrix} 1-p \\ p \end{pmatrix}$$
$$P_{\tilde{M}} = \begin{pmatrix} P(\tilde{M}=0) \\ P(\tilde{M}=1) \end{pmatrix} = \begin{pmatrix} 1-\tilde{p} \\ \tilde{p} \end{pmatrix}$$

$$\begin{cases} P(\tilde{M}=0) = P(\tilde{M}=0|M=0)P(M=0) + P(\tilde{M}=0|M=1)P(M=1) \\ P(\tilde{M}=1) = P(\tilde{M}=1|M=0)P(M=0) + P(\tilde{M}=1|M=1)P(M=1) \end{cases}$$

$\tilde{p} = \epsilon(1-p) + (1-\epsilon)p$

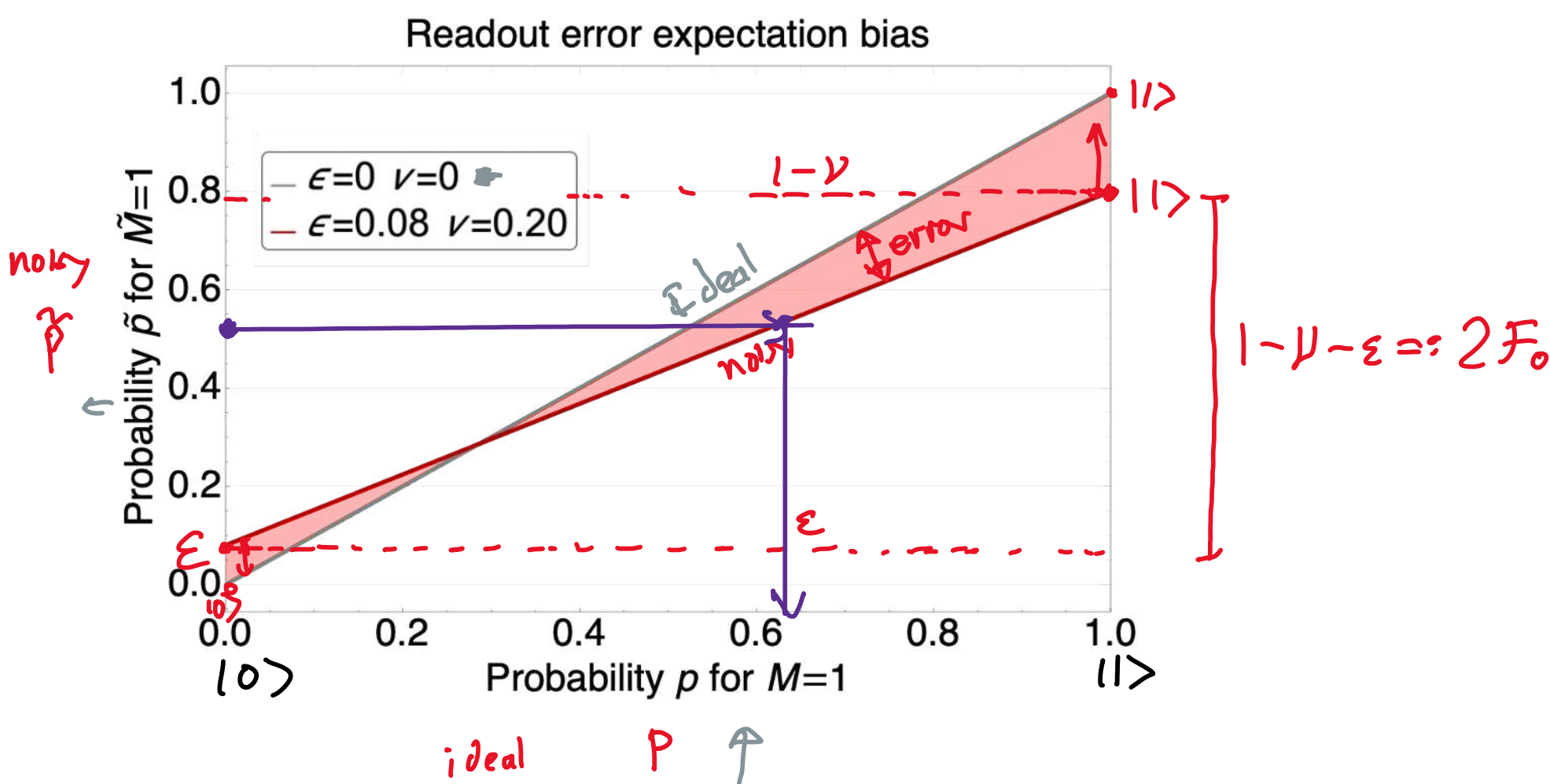
$$P_{\tilde{M}} = A P_M$$

$$A = \begin{pmatrix} P(\tilde{M}=0|M=0) & P(\tilde{M}=0|M=1) \\ P(\tilde{M}=1|M=0) & P(\tilde{M}=1|M=1) \end{pmatrix}$$
$$= \begin{pmatrix} 1-\epsilon & \epsilon \\ \epsilon & 1-\epsilon \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ for an ideal measurement}$$

$$\sum_n A_{nn} = 1 \text{ for any } n$$

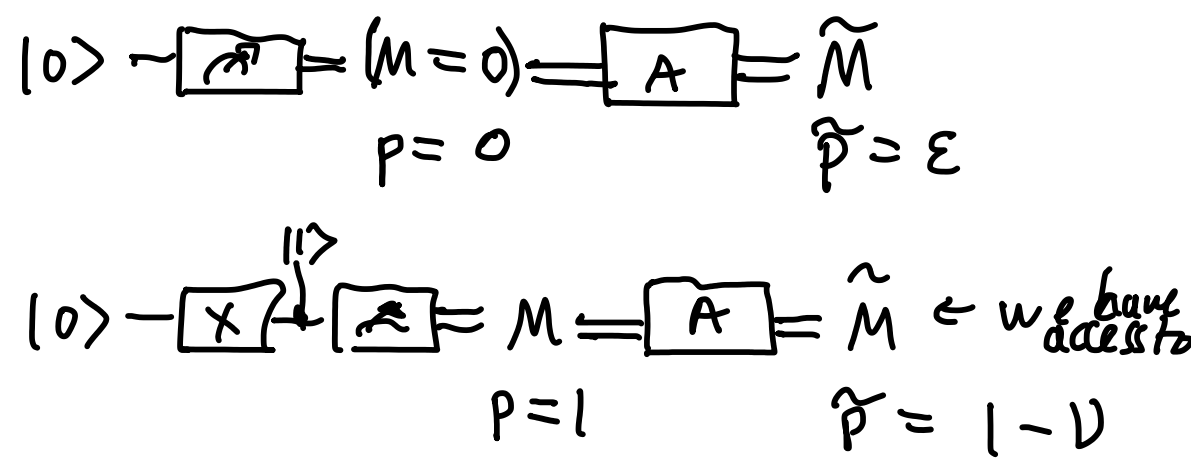
Stochastic matrix

$$\begin{aligned} \tilde{p} &= \epsilon(1-p) + (1-\epsilon)p \\ &= \epsilon - \epsilon p + p - \epsilon p \\ &= p + \epsilon - (2\epsilon)p \end{aligned}$$



Bonus section content:

Reconstruct A matrix



Noise mitigation

We know A

measure $\tilde{p}, \tilde{P}_M$ (noisy)

find p, P_M (ideal)

$$\tilde{P}_M = A P_M$$
$$P_M = A^{-1} \tilde{P}_M$$
$$p =$$

$\dim A = 2^n \times 2^n$ $n = \text{number of qubits}$

Assignment Fidelity

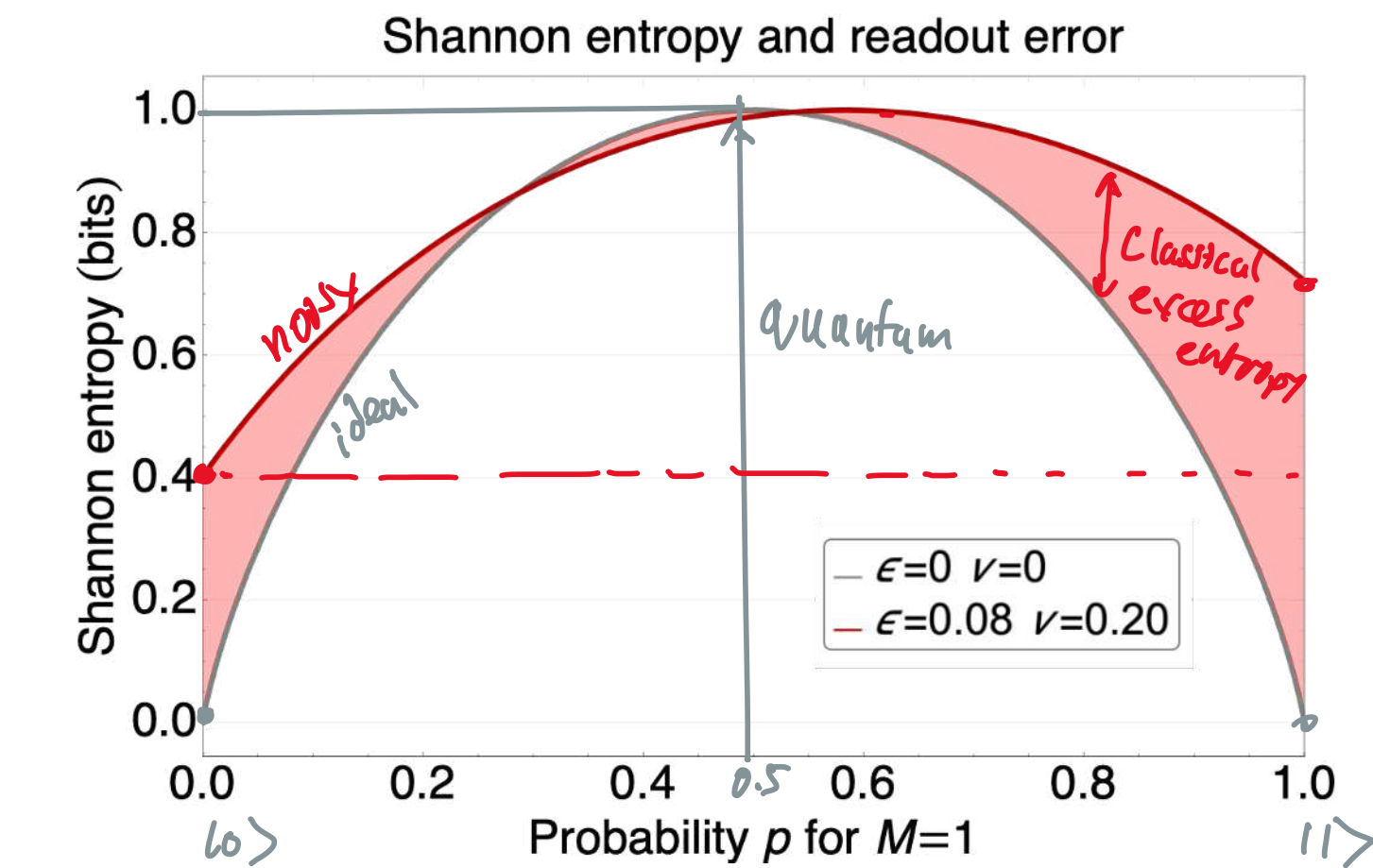
$$F_0 = 1 - \frac{1}{d} [P(\tilde{M}=1|M=0) + P(\tilde{M}=0|M=1)]$$
$$= \frac{1}{d} \text{Tr}(A)$$
$$= 1 - \frac{1}{d} (1-\nu)$$

$d = 2^n, n = \text{number of qubits}$

Shannon Entropy

$$H(A) = H(P_M) = - \sum_m P_m \log_2 P_m = - (1-p) \log_2 (1-p) - p \log_2 p$$

Binary entropy



Larger System

